σ-algebras, measures, and their properties

REAL VARIABLES I

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The preceding discussion of outer measure provides a size for every subset of X, but the Vitali counterexample shows that no countably additive, translation-invariant extension of interval length can be defined on every subset of $\mathbb{R}$. To obtain additivity, we must restrict the domain of a measure from 2^X to a carefully chosen collection of sets.

That collection must still support the standard operations consistens with measure properties: complements, countable unions, and therefore countable intersections. σ-algebras supply exactly this structure. Once the domain is in place, countable additivity yields the basic inequalities and continuity properties that make measures useful in later integration theory.

After studying these notes, you will be able to do the following.

- Distinguish rings, algebras, and σ-algebras and use their closure properties.
- Generate a σ-algebra and identify standard Borel generating families.
- Verify that a set function is a measure and derive its basic inequalities.
- Relate σ additivity, finite superadditivity, and monotone continuity.
- Compute measures of monotone limits of sets and recognize why the finite-measure hypothesis in continuity from above is necessary.

4.1 Algebras, sigma-algebras, and generators

The domain of a measure must be stable under the operations used in countable decompositions. Rings and algebras isolate the finite operations; σ-algebras add countable unions.

Definition 4.1 (Rings, algebras, and σ-algebras of sets). Let $\mathcal{A} \subseteq 2^X$ be a non-empty collection

- It is a *ring of sets* if

Empty set $\emptyset \in \mathcal{A}$.

Complement $A \in \mathcal{A}$ implies $X \setminus A \in \mathcal{A}$.

ion (and intersection) $A, B \in \mathcal{A}$ implies $A \cup B \in \mathcal{A}$ and $A \cap B \in \mathcal{A}$.

- The collection $\mathcal{A}$ is an *algebra of sets* if, in addition, we have

Entire set $X \in \mathcal{A}$.

- The collection $\mathcal{A}$ is a *σ-algebra* if, in addition

union and intersection Given $(A_n \in \mathcal{A})_{n\in\mathbb{N}}, B \in \mathcal{A}$, one has $\bigcup_{n\in\mathbb{N}} A_n$ and $\bigcap_{n\in\mathbb{N}} A_n$.

Note that once one shows closure with respect to complement and union, intersection follows by De Morgan's law: $A \cap B = (A^c \cup B^c)^c$.

Definition 4.2 (Measurable space). A *measurable space* is a pair $(X, \mathcal{A})$ in which X is a set and $\mathcal{A}$ is a σ-algebra on X. The elements of $\mathcal{A}$ are called the *measurable sets* of the space.

4.1.1 Examples and generated sigma-algebras

The collections $\{\emptyset, X\}$ and 2^X are σ-algebras on X. They are called the *trivial* and *discrete* σ-algebras, respectively.

A slightly more involved example is that of a σ-algebra generated by a partition.

Example 4.3 (σ-algebra of a finite partition). Let $P_1, \dots, P_N$ be nonempty finite partition of X, that is a pairwise disjoint colleciton of subsets of X such that $X = \bigcup_{j=1}^{N} P_j$. Then

$$\mathcal{A} \coloneqq \left\{ \bigcup_{j \in J} P_j : J \subseteq \{1, \dots, N\} \right\}$$

is a σ-algebra. Its elements are precisely the unions of cells of the partition.

Indeed, the empty union is $\emptyset$ and the union of all cells is X. The complement of the union indexed by J is the union indexed by $\{1, \dots, N\} \setminus J$. A countable union of elements of $\mathcal{A}$ is the union of the cells appearing in at least one of them, and is therefore again in $\mathcal{A}$.

More in general, given any collection of sets we define the generated σ-algebra as the "minimal" σ-sigma algebra containing those sets. Such a notion is valid because of the following property.

Proposition 4.4 (Intersections of σ-algebras). Let I be a nonempty index set, and let $(\mathcal{A}_i)_{i \in I}$ be σ-algebras on X. Then $\bigcap_{i \in I} \mathcal{A}_i$ is a σ-algebra on X.

Proof of Proposition 4.4.

- Every $\mathcal{A}_i$ contains X and $\emptyset$, so their intersection contains X.
- If A belongs to every $\mathcal{A}_i$, then A^c belongs to every $\mathcal{A}_i$, and therefore belongs to the intersection.
- If every A_n belongs to the intersection, then each $\mathcal{A}_i$ contains every A_n and hence contains $\bigcup_n A_n$. Thus the union belongs to the intersection.

□

Thanks to the property above, we can define the minimal σ-algebra of a set X that satisfies given properties. Indeed, one just takes the intersection of all σ-algebras of X satisying that property. The resulting object is a σ-algebra by the property above and is clearly not larger than any other σ-algebra satisfying the properties in question (in virtue of being the intersection).

Definition 4.5 (Generated σ-algebra). Let $\mathcal{G} \subseteq 2^X$. The *$\sigma$-algebra generated by $\mathcal{G}$* is

$$\sigma(\mathcal{G}) \coloneqq \bigcap \{ \mathcal{A} \subseteq 2^X : \mathcal{A} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{G} \subseteq \mathcal{A} \}.$$

The collection being intersected is nonempty because 2^X is a σ-algebra containing $\mathcal{G}$. By Proposition 4.4, $\sigma(\mathcal{G})$ is a σ-algebra. It contains $\mathcal{G}$, and it is contained in every σ-algebra that contains $\mathcal{G}$. These facts characterize $\sigma(\mathcal{G})$ as the smallest σ-algebra containing $\mathcal{G}$.

Generated σ-algebras are usually handled through this minimality property. For example, to compare two generating families, it suffices to show that each member of one family belongs to the σ-algebra generated by the other. An explicit list of all elements of $\sigma(\mathcal{G})$ is, on the other hand, very difficult to provide.

4.1.2 Borel sigma-algebras

Definition 4.6 (Borel σ-algebra). Let $(X, \mathcal{T})$ be a topological space. The *Borel σ-algebra* of X is $\mathcal{B}(X) \coloneqq \sigma(\mathcal{T})$, the σ-algebra generated by the open subsets of X. Its elements are the *Borel sets* of X.

On Euclidean space, a countable family of rectangles generates all Borel sets.

Proposition 4.7 (Rational open cubes generate the Euclidean Borel sets). Let $d \in \mathbb{N}$ with $d \geq 1$, and let $Q_r(c) \coloneqq c + (-r, r)^d$. The familty $\mathcal{Q} \coloneqq \{Q_r(c) : r \in \mathbb{Q},\, c \in \mathbb{Q}^d,\, r > 0\}$ is countable and it generates the Borel σ-algebra i.e. $\mathcal{B}(\mathbb{R}^d) = \sigma(\mathcal{Q})$.

Proof. The family $\mathcal{Q}$ is countable because it is parametrized by $\mathbb{Q}^d \times (\mathbb{Q} \cap (0, \infty))$. Since every member of $\mathcal{Q}$ is open, $\sigma(\mathcal{Q}) \subseteq \mathcal{B}(\mathbb{R}^d)$. It remains to show that every open set belongs to $\sigma(\mathcal{Q})$.
Let $A \subseteq \mathbb{R}^d$ be open. For $z \in A$, define $\bar{r}_z \coloneqq \sup\{r > 0 : Q_r(z) \subseteq A\}$. Then $\bar{r}_z > 0$. For each $c \in A \cap \mathbb{Q}^d$, choose $r'_c \in \mathbb{Q} \cap (0, \infty)$ such that $Q_{r'_c}(c) \subseteq A$, with $r'_c \in (\bar{r}_c/2, \bar{r}_c)$ when $\bar{r}_c < \infty$, and with $r'_c > 1$ when $\bar{r}_c = \infty$. We claim that

$$A = \bigcup_{c \in A \cap \mathbb{Q}^d} Q_{r'_c}(c).$$

The right-hand side is contained in A by construction. Conversely, fix $z \in A$, and choose $0 < t \leq 1$ with $Q_t(z) \subseteq A$. Density of $\mathbb{Q}^d$ gives $c \in \mathbb{Q}^d$ with $\|z - c\|_\infty < t/8$. Then $c \in A$ and $\bar{r}_c > 7t/8$. If $\bar{r}_c < \infty$, then $r'_c > 7t/16 > \|z - c\|_\infty$; if $\bar{r}_c = \infty$, then $r'_c > 1 > \|z - c\|_\infty$. Thus $z \in Q_{r'_c}(c)$, proving the claim.
The displayed union is countable, so $A \in \sigma(\mathcal{Q})$. Hence $\mathcal{B}(\mathbb{R}^d) \subseteq \sigma(\mathcal{Q})$, and the reverse inclusion above gives the result. □

The choice of open rational rectangles is convenient but not essential.

Problem 4.8 (Equivalent rectangle generators). Let $d \in \mathbb{N}$ with $d \geq 1$. Let $\mathcal{R}_{\text{op}}$, $\mathcal{R}_{\text{cl}}$, and $\mathcal{R}_{\text{ho}}$ denote, respectively, the collections of bounded axis-parallel rectangles of the forms

$$\prod_{j=1}^{d} (a_j, b_j), \qquad \prod_{j=1}^{d} [a_j, b_j], \qquad \prod_{j=1}^{d} [a_j, b_j),$$

where $a_j < b_j$ are real numbers. Then, show that $\sigma(\mathcal{R}_{\text{op}}) = \sigma(\mathcal{R}_{\text{cl}}) = \sigma(\mathcal{R}_{\text{ho}}) = \mathcal{B}(\mathbb{R}^d)$. Show that the same conclusion holds if every endpoint is required to be rational.

4.2 Measures and their basic properties

A measure assigns a nonnegative extended real number to every set in a σ-algebra.

Definition 4.9 (Measure and measure space). Let $(X, \mathcal{A})$ be a measurable space. A *measure* on $(X, \mathcal{A})$ is a function $\mu \colon \mathcal{A} \longrightarrow [0, +\infty]$ such that

1. $\mu(\emptyset) = 0$;
2. for every pairwise disjoint sequence $(E_n \in \mathcal{A})_{n \in \mathbb{N}}$,

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n).$$

The second property is *countable additivity* or *σ-additivity*. A triple $(X, \mathcal{A}, \mu)$ is a *measure space*.

Countable additivity and nonnegativity ensure that every measure is monotone and countably subadditive. Consequently, on its σ-algebra, a measure satisfies all defining properties of an outer measure.

Indeed, if $A, B \in \mathcal{A}$ satisfy $A \subseteq B$, then $B = A \cup (B \setminus A)$ decomposes B into disjoint measurable sets. Countable additivity gives

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A),$$

where the inequality follows from nonnegativity of $\mu(B \setminus A)$. This establishes monotonicity.

To establish countable subadditivity, let $(A_n \in \mathcal{A})_{n\in\mathbb{N}}$ be an arbitrary sequence of measurable sets. First form the increasing sequence $U_n \coloneqq \bigcup_{k=1}^{n} A_k$, and then represent its union as the disjoint union $\bigcup_{n=1}^{\infty} \tilde{A}_n$, where

$$\tilde{A}_1 \coloneqq A_1 \qquad \text{and} \qquad \tilde{A}_n \coloneqq U_n \setminus U_{n-1} = A_n \setminus \left(\bigcup_{k=1}^{n-1} A_k \right) \quad (n \geq 2).$$

Each $\tilde{A}_n$ belongs to $\mathcal{A}$, the sets $(\tilde{A}_n)_{n\in\mathbb{N}}$ are pairwise disjoint by construction, and each satisfies $\tilde{A}_n \subseteq A_n$. Countable additivity and monotonicity then yield

$$\mu\left(\bigcup_{n=1}^{\infty} A_n \right) = \mu\left(\bigcup_{n=1}^{\infty} \tilde{A}_n \right) = \sum_{n=1}^{\infty} \mu(\tilde{A}_n) \leq \sum_{n=1}^{\infty} \mu(A_n),$$

which proves σ-subadditivity.

4.2.1 Examples and size conditions

Example 4.10 (Weighted counting measures). Let $w\colon X \to [0, +\infty]$. For $E \subseteq X$, define

$$\mu_w(E) \coloneqq \sum_{x\in E} w(x),$$

where the nonnegative sum is the supremum of its finite subsums. Then μ_w is a measure on $(X, 2^X)$.

The empty sum is zero. If $(E_n)_{n\in\mathbb{N}}$ is pairwise disjoint, then every $x \in \bigcup_n E_n$ occurs in exactly one of the sets. Reindexing the nonnegative sum gives

$$\mu_w\left(\bigcup_{n=1}^{\infty} E_n \right) = \sum_{x\in\bigcup_n E_n} w(x) = \sum_{n=1}^{\infty} \sum_{x\in E_n} w(x) = \sum_{n=1}^{\infty} \mu_w(E_n).$$

The definition by finite subsums makes this reindexing valid whether the value is finite or infinite. Taking $w(x) = 1$ for every x gives *counting measure*:

$$\#E = \begin{cases} \text{the number of elements of } E, & E \text{ finite}, \\ +\infty, & E \text{ infinite}. \end{cases}$$

Example 4.11 (Dirac point mass). Let $(X, \mathcal{A})$ be a measurable space and fix $x_0 \in X$. The *Dirac measure at* x_0 is

$$\delta_{x_0}(E) \coloneqq \begin{cases} 1, & x_0 \in E, \\ 0, & x_0 \notin E, \end{cases} \qquad E \in \mathcal{A}.$$

We have $\delta_{x_0}(\emptyset) = 0$. For a pairwise disjoint sequence, at most one set contains x_0. Both sides of the countable-additivity identity are therefore 1 if x_0 belongs to the union and 0 otherwise.

Example 4.12 (Anticipated Lebesgue measure). For a positive integer d, we will later construct a σ-algebra $\mathcal{L}(\mathbb{R}^d)$ containing $\mathcal{B}(\mathbb{R}^d)$ and a measure

$$\lambda\colon \mathcal{L}(\mathbb{R}^d) \longrightarrow [0, +\infty]$$

such that $\lambda(R)$ is the Euclidean volume of every axis-parallel rectangle R. This measure is Lebesgue measure. Its construction from Lebesgue outer measure is postponed.

Definition 4.13 (Finite, probability, and σ-finite measures). Let $(X, \mathcal{A}, \mu)$ be a measure space.

1. The measure μ is *finite* if $\mu(X) < +\infty$.
2. The measure μ is a *probability measure* if $\mu(X) = 1$.
3. The measure μ is *σ-finite* if there is a sequence $(X_n \in \mathcal{A})_{n\in\mathbb{N}}$ such that

$$X = \bigcup_{n=1}^{\infty} X_n \qquad \text{and} \qquad \mu(X_n) < +\infty \quad \text{for every } n.$$

Every probability measure is finite, and every finite measure is σ-finite by taking $X_n = X$. The converses fail. Counting measure on $\mathbb{N}$ is σ-finite because $\mathbb{N}$ is a countable union of singletons, but it is not finite. Counting measure on an uncountable set X is not σ-finite: every set of finite counting measure is finite, while a countable union of finite sets is countable. The Dirac measure δ_{x_0} is a probability measure. On $X = \{1, 2, 3, ...\}$, the weighted counting measure with $w(n) = 2^{-n}$ is also a probability measure.

Conditional on the construction in Example 4.12, Lebesgue measure on $\mathbb{R}^d$ is σ-finite because the boxes $[-n, n]^d$ have finite measure and cover $\mathbb{R}^d$. It is not finite: monotonicity and the rectangle formula give

$$\lambda(\mathbb{R}^d) \geq \lambda([-n, n]^d) = (2n)^d$$

for every $n \geq 1$.

4.2.2 Consequences of countable additivity

Countable additivity determines how a measure interacts with finite operations and monotone limits. We write $E_n \uparrow E$ when $E_n \subseteq E_{n+1}$ for every n and $E = \bigcup_{n=1}^{\infty} E_n$.

Proposition 4.14 (Basic properties of measures). Let $(X, \mathcal{A}, \mu)$ be a measure space. The following properties hold.

Finite additivity If $E_1, \dots, E_N \in \mathcal{A}$ are pairwise disjoint, then

$$\mu\left(\bigcup_{n=1}^{N} E_n\right) = \sum_{n=1}^{N} \mu(E_n).$$

Monotonicity If $A, B \in \mathcal{A}$ and $A \subseteq B$, then $\mu(A) \leq \mu(B)$.

ifference formula If $A, B \in \mathcal{A}$, $A \subseteq B$, and $\mu(A) < +\infty$, then

$$\mu(B \setminus A) = \mu(B) - \mu(A).$$

(σ-subadditivity) For every sequence $(A_n \in \mathcal{A})_{n\in\mathbb{N}}$,

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n).$$

-superadditivity) If $(E_n \in \mathcal{A})_{n\in\mathbb{N}}$ is pairwise disjoint and $E \in \mathcal{A}$ satisfies $\bigcup_{n=1}^{\infty} E_n \subseteq E$, then

$$\mu(E) \geq \sum_{n=1}^{\infty} \mu(E_n).$$

nuity from below If $(E_n \in \mathcal{A})_{n\in\mathbb{N}}$ and $E_n \uparrow E$, then

$$\mu(E_n) \uparrow \mu(E).$$

Proof of Proposition 4.14. Step 1: reduce finite additivity to countable additivity.
For a finite sequence of pairwise disjoint sets $E_1, \dots, E_N \in \mathcal{A}$, define $E_n = \emptyset$ for each $n > N$. Because $\mu(\emptyset) = 0$, countable additivity gives

$$\mu\left(\bigcup_{n=1}^{N} E_n\right) = \mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n) = \sum_{n=1}^{N} \mu(E_n).$$

Step 2: decompose a set difference and verify monotonicity.
If $A \subseteq B$, the sets A and $B \setminus A$ are disjoint and satisfy $B = A \dot\cup (B \setminus A)$. Finite additivity gives

$$\mu(B) = \mu(A) + \mu(B \setminus A).$$

Because $\mu(B \setminus A) \geq 0$, we obtain $\mu(A) \leq \mu(B)$. This recovers the monotonicity property established in Section 4.2. When $\mu(A) < +\infty$, subtracting the finite quantity $\mu(A)$ from both sides gives the difference formula $\mu(B \setminus A) = \mu(B) - \mu(A)$.

Step 3: subadditivity and superadditivity.
Countable subadditivity was established following Definition 4.9 in Section 4.2. For countable superadditivity, if $(E_n \in \mathcal{A})_{n\in\mathbb{N}}$ is pairwise disjoint and $\bigcup_{n=1}^{\infty} E_n \subseteq E$, then monotonicity and countable additivity immediately yield

$$\mu(E) \geq \mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n).$$

Step 4: continuity along increasing sequences.
Suppose $E_n \uparrow E$. Define the disjoint increments

$$D_1 \coloneqq E_1 \qquad \text{and} \qquad D_n \coloneqq E_n \setminus E_{n-1} \quad (n \geq 2).$$

The sets $(D_n)_{n\in\mathbb{N}}$ are measurable and pairwise disjoint, with

$$E_N = \bigcup_{n=1}^{N} D_n \qquad \text{and} \qquad E = \bigcup_{n=1}^{\infty} D_n.$$

Finite additivity and countable additivity give

$$\mu(E_N) = \sum_{n=1}^{N} \mu(D_n) \qquad \text{and} \qquad \mu(E) = \sum_{n=1}^{\infty} \mu(D_n).$$

Because the partial sums of a series with nonnegative terms increase to the infinite sum, we conclude that $\mu(E_N) \uparrow \mu(E)$. □

Remark 4.15 (Three useful sequence transformations). The following conversions let us move between the hypotheses of the main measure identities.

- If $E_n \uparrow E$, its *disjoint increments* are $D_1 = E_1$ and $D_n = E_n \setminus E_{n-1}$ for $n \geq 2$. Then $E_N = \bigcup_{n=1}^{N} D_n$ and $E = \bigcup_{n=1}^{\infty} D_n$. This is the key step in the proof of continuity from below in Proposition 4.14.
- If (A_n) is pairwise disjoint, its finite unions $S_N = \bigcup_{n=1}^{N} A_n$ form an increasing sequence with the same total union. This is used to recover countable additivity from continuity from below in Proposition 4.19.
- For an arbitrary sequence (A_n), first take the finite unions $U_n = \bigcup_{k=1}^{n} A_k$, and then take the disjoint increments U_1 and $U_n \setminus U_{n-1}$. This is the disjointification used for countable subadditivity in Proposition 4.14.

Warning (Cancellation at infinity). The difference formula cannot be obtained by formal cancellation when $\mu(A) = +\infty$. For counting measure on $X = \mathbb{N} \cup \{p\}$, take $A = \mathbb{N}$ and $B = X$. Then $\mu(B \setminus A) = 1$, while the expression $\mu(B) - \mu(A) = +\infty - (+\infty)$ is undefined.

The two-set form of inclusion–exclusion can be written without any subtraction:

$$\mu(A \cup B) + \mu(A \cap B) = \mu(A) + \mu(B).$$

Indeed, after decomposing $A \cup B$ into the three disjoint sets $A \setminus B$, $A \cap B$, and $B \setminus A$, both sides equal

$$\mu(A \setminus B) + \mu(B \setminus A) + 2\mu(A \cap B).$$

The identity remains meaningful when one or both sides are infinite.

Corollary 4.16 (Finite inclusion–exclusion). Let $A_1, \dots, A_N \in \mathcal{A}$, and assume $\mu(A_j) < +\infty$ for every j. Then every term below is finite and

$$\mu\left(\bigcup_{j=1}^{N} A_j\right) = \sum_{\emptyset \neq J \subseteq \{1,\dots,N\}} (-1)^{|J|+1} \mu\left(\bigcap_{j \in J} A_j\right).$$

Proof of Corollary 4.16. Every nonempty intersection in the formula is contained in one of the sets A_j, so its measure is finite by monotonicity. The formula for $N = 2$ follows by subtracting the finite value $\mu(A_1 \cap A_2)$ from the two-set identity. Induction on N, applying the two-set formula to $\bigcup_{j=1}^{N-1} A_j$ and A_N, gives the displayed identity. □

We write $E_n \downarrow E$ when $E_{n+1} \subseteq E_n$ for every n and $E = \bigcap_{n=1}^{\infty} E_n$. While continuity along increasing sequences holds unconditionally (Proposition 4.14), continuity along decreasing sequences requires a finiteness hypothesis.

Proposition 4.17 (Continuity from above). Let $(X, \mathcal{A}, \mu)$ be a measure space. If $(E_n \in \mathcal{A})_{n \in \mathbb{N}}$, $E_n \downarrow E$, and $\mu(E_1) < +\infty$, then

$$\mu(E_n) \downarrow \mu(E).$$

Proof of Proposition 4.17. Set $F_n = E_1 \setminus E_n$. Then $F_n \uparrow E_1 \setminus E$. By continuity from below (Proposition 4.14),

$$\mu(F_n) \uparrow \mu(E_1 \setminus E).$$

Since $\mu(E_1) < +\infty$, all terms are finite and $\mu(E_n) = \mu(E_1) - \mu(F_n)$. Taking limits gives

$$\lim_{n\to\infty} \mu(E_n) = \mu(E_1) - \mu(E_1 \setminus E) = \mu(E).$$

□

Example 4.18 (Failure of continuity from above without finite measure). Let $X = \mathbb{N}$, let $\mathcal{A} = \mathcal{P}(\mathbb{N})$, and let μ be counting measure. For $E_n = \{n, n+1, n+2, ...\}$, we have $E_n \downarrow \emptyset$, but $\mu(E_n) = +\infty$ for every n while $\mu(\emptyset) = 0$. Thus continuity from above can fail when $\mu(E_1) = +\infty$.

4.2.3 Finite superadditivity and a characterization of measures

Every nonnegative, finitely additive set function is automatically countably superadditive on disjoint sets. Indeed, let $\nu\colon \mathcal{A} \to [0, +\infty]$ be finitely additive with $\nu(\emptyset) = 0$. If $A, B \in \mathcal{A}$ satisfy $A \subseteq B$, then the decomposition $B = A \dot\cup (B \setminus A)$ gives $\nu(B) = \nu(A) + \nu(B \setminus A) \geq \nu(A)$, so ν is monotone. If $(E_n \in \mathcal{A})_{n\in\mathbb{N}}$ is pairwise disjoint and $E \in \mathcal{A}$ satisfies $\bigcup_{n=1}^{\infty} E_n \subseteq E$, monotonicity and finite additivity imply

$$\nu(E) \geq \nu\left(\bigcup_{n=1}^{N} E_n\right) = \sum_{n=1}^{N} \nu(E_n)$$

for every $N \geq 1$. Taking the limit as $N \to \infty$ gives $\nu(E) \geq \sum_{n=1}^{\infty} \nu(E_n)$. In particular, setting $E = \bigcup_{n=1}^{\infty} E_n$ whenever this union belongs to $\mathcal{A}$ yields countable superadditivity:

$$\nu\left(\bigcup_{n=1}^{\infty} E_n\right) \geq \sum_{n=1}^{\infty} \nu(E_n).$$

Because a finitely additive set function already satisfies this inequality, establishing that ν is a measure requires only the reverse inequality. Continuity along increasing sequences provides precisely this bridge.

Proposition 4.19 (Countable additivity from continuity below). Let $(X, \mathcal{A})$ be a measurable space, and let $\nu\colon \mathcal{A} \longrightarrow [0, +\infty]$ satisfy $\nu(\emptyset) = 0$ and finite additivity on disjoint sets of $\mathcal{A}$. Then ν is a measure if and only if it is continuous from below:

$$\nu(E_n) \uparrow \nu(E) \text{ for any } (E_n \in \mathcal{A})_{n\in\mathbb{N}} \text{ with } E_n \uparrow E.$$

Proof of Proposition 4.19. Step 1: obtain continuity from a measure.
If ν is a measure, continuity from below is item 6 of Proposition 4.14.

Step 2: recover countable additivity from continuity.
Conversely, suppose that ν is continuous from below, and let $(A_n \in \mathcal{A})_{n\in\mathbb{N}}$ be pairwise disjoint. Define

$$S_N = \bigcup_{n=1}^{N} A_n, \qquad S = \bigcup_{n=1}^{\infty} A_n.$$

Then $S_N \uparrow S$. Finite additivity and continuity from below give

$$\nu(S) = \lim_{N\to\infty} \nu(S_N) = \lim_{N\to\infty} \sum_{n=1}^{N} \nu(A_n) = \sum_{n=1}^{\infty} \nu(A_n).$$

Thus ν is countably additive. □